A graphon does not determine a first-order limit

Disproof of Conjecture 00000009183

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Abstract

The complete graphs and the complete graphs with one edge deleted have the same constant-one graphon limit. Both sequences have first-order limits, but the sentence asserting completeness distinguishes them at every size. Thus graphons cannot characterize first-order limits of dense graphs in the sense of determining their limiting first-order theories. The ordinary dense graph limit theorem is unaffected.

Statement and interpretation

The English source begins "Definition: First-order limits of dense graphs" and describes graphons as the characterization of the limit objects. The Chinese source likewise explicitly says first-order limits. We test the following necessary consequence of a graphon characterization: if two first-order convergent dense graph sequences have the same graphon limit, they have the same eventual truth value for every first-order sentence in the language with adjacency and equality.

The source does not specify a formal convergence definition. We use the usual first-order convergence in which the proportions of satisfying assignments converge for every formula; for a sentence these proportions are its truth values, zero or one. Our sequences are convergent even in this stronger formula sense, as shown below. If "first-order" in the source were removed and only convergence of homomorphism densities were intended, the familiar graphon characterization would be correct. The counterexample concerns the first-order assertion actually present in both versions, not that corrected assertion.

The counterexample

For every $N \geq 2$, let $A_N = K_N$, and obtain B_N by deleting exactly one edge, whose endpoints are denoted u, v. Both graphs have N vertices. Their ordered edge densities are

$$\frac{N(N-1)}{N^2} \longrightarrow 1, \qquad \frac{N(N-1)-2}{N^2} \longrightarrow 1.$$

In particular both sequences are dense.

Partition $[0,1)$ into N intervals I_i of length $1/N$, and let W_G be the usual adjacency step graphon: on $I_i \times I_j$ it is 1 if i, j are adjacent and 0 otherwise. Values at the endpoint 1 are immaterial. There are N diagonal zero squares for A_N, and $N+2$ zero squares for B_N. Consequently

$$\|1 - W_{A_N}\|_1 = \frac{1}{N}, \qquad \|1 - W_{B_N}\|_1 = \frac{N+2}{N^2}.$$

The cut norm is bounded above by the L^1 norm, so both graphons converge in cut norm, and hence in cut distance, to the same graphon $W \equiv 1$. This includes the diagonal error; it is not merely a comparison of the two edge counts.

For completeness the same conclusion can be checked against every finite test graph. If F has m edges, independent uniform choices of its vertex images give, by the union bound,

$$0 \le 1 - t(F, A_N) \le \frac{m}{N}, \qquad 0 \le 1 - t(F, B_N) \le \frac{m(N+2)}{N^2}.$$

Each test edge fails with the displayed missing ordered-pair probability; independence between the different failure events is not needed. Thus all finite homomorphism densities converge to those of $W = 1$.

Now consider the first-order sentence

$$\sigma := \forall x \, \forall y \, (x \neq y \implies E(x, y)).$$

Every A_N satisfies σ, whereas every B_N fails it, witnessed by u, v. Their limiting first-order theories therefore differ.

Existence of the first-order limits

Here we also verify that the example is not relying on a sequence which fails to have a first-order limit. For A_N, A_M, the duplicator wins the q-round Ehrenfeucht–Fraisse game whenever $N, M \ge q$: equality of already selected vertices is preserved, and a fresh vertex is matched by a fresh vertex. All distinct selected vertices are adjacent.

For B_N, B_M, the duplicator preserves the pair of exceptional vertices $\{u, v\}$, matching exceptional vertices to exceptional vertices and ordinary vertices to ordinary vertices. Once an exceptional vertex is selected its match is fixed, and the other exceptional vertex is matched to the remaining exceptional vertex if selected. Fresh ordinary vertices are always available when $N-2, M-2 \ge q$. This is a partial isomorphism after every move: the only missing edge between distinct vertices is the exceptional pair. Thus every sentence of quantifier rank at most q has constant truth value on each sequence once $N \ge q + 2$.

For the version of first-order convergence including formulas with free variables, fix a formula with k free variables and quantifier rank q. The proportion of assignments with a repeated vertex or an exceptional vertex is at most

$$\binom{k}{2}/N + 2k/N,$$

which tends to zero. (For A_N, omit the second term.) All remaining assignments are injective and ordinary. Any two such assignments, in graphs of sizes at least $k + q + 2$, extend the same partial isomorphism in the q-round game above. The formula therefore has one fixed truth value on these assignments, independent of the sufficiently large size. Its satisfaction proportion converges to that value. This proves the claimed first-order convergence for every formula.

Formal verification and scope

The Lean development uses Mathlib's `SimpleGraph` on $\mathrm{Fin}(n+2)$, so $N = n+2$. It defines both graph sequences, their actual zero–one adjacency kernels, and the explicit semantic property represented by σ. It proves the truth and falsity of that property on the two graphs for every n.

To connect the finite kernels to cut convergence, let $a_i, b_j \in [0, 1]$ be arbitrary. Lean proves nonnegativity and the uniform bound

$$0 \le \frac{1}{N^2} \sum_{i,j} a_i b_j (1 - W_{B_N}(i, j)) \le \frac{N+2}{N^2},$$

and the same upper bound for A_N, together with convergence of the upper bound to zero. This quantifies over all fractional rectangle weights, not only subsets of the vertex set. For measurable S, T, put $a_i = N\lambda(S \cap I_i)$ and $b_j = N\lambda(T \cap I_j)$. The left side is exactly $\int_{S\times T}(1 - W_G)$ by the elementary integral formula for step functions. Since $1 - W_G \geq 0$, these inequalities are precisely uniform control of all cut rectangles.

The development defines this finite-kernel cut convergence and the necessary assertion that a common constant-one limit determines the eventual truth of σ, and proves its negation. The final theorem is

`Conjecture9183.conjecture9183_false`.

Its statement is the negation of

`GraphonDeterminesCompleteness`.

The only axioms reported are `propext`, `Classical.choice`, and `Quot.sound`.

The identification of the finite weighted sums with measurable step-graphon integrals, the homomorphism-density union bound, and the Ehrenfeucht–Fraisse arguments are proved here on paper. Lean does not implement a general syntax or satisfaction relation for arbitrary first-order formulas, or a general graphon measure space. Its sentence predicate is exactly the semantics of the displayed sentence. The source is informal; the Lean proposition explicitly records the necessary consequence being refuted instead of claiming a literal formalization of undefined prose.

Source and reproduction

The unmodified bilingual statement is supplied as `SOURCE.md`. The source was frozen by the coordinating workflow; its SHA-256 is in the provenance file. Lean is pinned to 4.19.0 and Mathlib commit

`c44e0c8ee63ca166450922a373c7409c5d26b00b`.

From the supplied Lean project, run

```
lake env lean -DwarningAsError=true Main.lean
```

This draft was developed by a delegated AI agent with a self-review; the coordinating agent reviews it separately. No external independent review is claimed.